fully ionized hot spot, the plasma is optically thick over a large wavelength range. Hence, all criteria for blackbody emission are fulfilled. Adiabatic cooling during the plasma expansion leads to the red shift in the periphery of the blurred region around the hot spot.

During ns breakdown at large *NA*, a hot plasma core is formed within the laser cone angle, which emits energetic photons producing a cascade of secondary electrons in a larger volume. These electrons then emit the diffuse blue luminescence visible on the photographs in Figs. 7(b) and (d). The luminescent halo exhibits no red shift in its periphery and is distributed non-isotropically around the plasma core, with largest extent in the propagation direction of the laser beam. Therefore, it cannot be related to the hydrodynamic plasma expansion. The primary radiation emitted by the high-density strings in the plasma core is thermal bremsstrahlung since the interplay of AI ($\propto \rho_c$), recombination ($\propto \rho_c^2$) and thermal ionization [$\propto exp\ (-E_{gap}/2k_BT)$] rapidly establishes a Maxwell Boltzmann distribution typical for LTE. However, it is not blackbody radiation because the strings are optically thin. By contrast, the diffuse luminescence of the inflated plasma fulfils the blackbody criteria, because it emerges from a fairly large region with high electron density such that the plasma is optically thick.

In this paper, we have determined the thermodynamic plasma temperature and the plasma pressure through an evaluation of the measured plasma energy density based on EOS data (Table 3). Alternatively, the plasma temperature can be determined by fitting the bremsstrahlung or blackbody emission spectrum to a Planckian distribution. This has been performed for brightly luminescent ns and ps plasmas [224-226]. In ns plasmas, peak plasma temperature values in strings and hot spots will significantly exceed the average values. However, such transient local temperature variations cannot be assessed by spatially and time-integrated spectroscopic measurements at $\lambda > 300$ nm, which detect the long-wavelength tail of the diffuse luminescence from the entire halo [224-226]. Fits of a Planckian distribution to plasma emission spectra from IR breakdown in water yielded blackbody temperatures of 6200 K for 80-ps, 1-mJ pulses, 10000 K for 5-ns, 4 mJ pulses [224,246], and values of 15000 to 16000 K at much larger pulse energies ≥ 100 mJ [225,226]. A value of 11500 K was reported for ns breakdown in fused silica [297]. Since the energy density of thermal radiation is proportional to T^4 [291,294], spectroscopic results will be close to the peak temperature within the depth from which radiation exits the luminescent halo. In inhomogeneous plasmas exhibiting a hot spot, they represent the temperature of the hot region rather than the average temperature over the entire plasma volume. This explains, why blackbody temperatures reported in the literature are higher than the average values in Table 3. These intricacies need to be

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considered, when plasma temperature and temperature data are used as input for hydrodynamics calculations of shock wave emission and cavitation bubble generation.

Recently, Tian et al. determined the temperature evolution during the early plasma and bubble expansion phase after optical breakdown in bulk water by 1064-nm, 10-ns pulses of 20 mJ energy during a time interval from 75 ns to 700 ns after the laser pulse [103]. For this purpose, line emission from OH radicals and Ca tracers dissolved in the water was measured and temperatures determined by fitting experimental OH spectra with synthetic spectra and through Saha-Boltzmann plots of calcium lines [299]. Temperature values at t = 75 ns were 12700 K from the Saha-Boltzmann plot and 11000 K from OH line spectra. The subsequent temperature decay from 75 ns to 700 ns matched well with adiabatic conditions during the early expansion phase, as expected from hydrodynamic bubble models [42,63]. However, the finding of a plasma temperature at t = 75 ns similar to the blackbody temperature right after the laser pulse in [224-226] is very remarkable. Presumably, a high temperature is maintained during the early plasma expansion and decay through energy release by electron-ion recombination and reformation of radicals and molecules from hydrogen and oxygen atoms. Such energy release resembles the release of latent heat during condensation which keeps the temperature constant during the phase change. However, the ionization and molecular dissociation energies are much larger than the latent heat of vaporization per water molecule (0.422 eV) and the effect is, hence, also much larger.

5.6. Energy partitioning in dependence on plasma energy density

The energy balance of optical breakdown comprises the fraction of incident laser energy that is absorbed and the partition of E_{abs} into E_v, E_B, and E_{SW}. The energy of the plasma radiation emitted from the diffusely luminescent plasma is here neglected because it amounts to less than 1% of E_{abs} [114,297] even though the radiative transport within the plasma involves a larger energy fraction. While the dependence of nonlinear energy deposition on pulse duration, wavelength, focusing angle and pulse energy is very complex as discussed in the previous sections, the partitioning of E_{abs} relies mainly on average plasma energy density and is governed by more general rules. In the following, we discuss the changes of partitioning with growing U_{avg}.

Below bubble threshold, all energy goes into free-electron mediated chemical effects and heating [22]. Numerical simulations for plasma formation slightly above the bubble threshold (R_{max} = 230 nm) yielded the result that 0.46% of the absorbed energy is transformed into the acoustic energy of the laser-produced thermoelastic transient, and only 0.03% goes into bubble energy [22]. This is consistent with

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the conversion efficiency data of Fig. 10, whereby these E_B/E_L data relate to the incident laser energy while modeling referred to E_B/E_{abs}.

There is an information gap on energy partitioning for U_{avg}-values close to threshold, where plasma luminescence is too weak to delineate the plasma shape. The plasma volume could be inferred from shadowgraphs or Schlieren pictures taken immediately after breakdown, or via third harmonic (THG) imaging [218,220] but such techniques were not applied in the present study. Furthermore, the determination of E_{abs} merely from transmission measurements through Eq. (9) becomes inaccurate, and one would need to measure also the amount of scattered light. Determination of V_{plasma} and E_{abs} close to threshold remains a task for future work.

Just above threshold, absorbed energy will partition mainly into vaporization energy and bubble energy, while little energy goes into shock wave emission. Our investigations do not cover this regime because energy partitioning can be evaluated only when plasma luminescence starts to be detectable on photographs, which in our setup is the case for $U_{avg} \geq 8.5$ kJ/cm^3 (the limit could be shifted by integrating over more images or use of an ICCD camera). Here, 30% of E_{abs} go into vaporization, almost 20% is converted into bubble energy, and about 50% goes into shock wave energy (Fig. 11). The work fractions needed to overcome surface tension and viscosity are negligible because luminescent-plasmas produce relatively large bubbles, for which the respective pressure terms scaling with $1/R$ are very small [42]. With further increase of plasma energy density, a decreasing part of E_{abs} is needed for vaporization and an ever-larger fraction can be converted into mechanical energy. The increase of the conversion into mechanical energy is rapid up to about 15 kJ/cm^3 and becomes more gradual for larger plasma energy densities. The fraction E_V/E_{abs} drops to only 7% at $\bar{\varepsilon}_{avg}$ = 40 kJ/cm^3, while E_B/E_{abs} stays approximately constant near 20% and E_{SW}/E_{abs} increases to 78%.

The ratio of shock wave energy and bubble energy, E_{SW}/E_B, increases from a value of ≈ 2.8 at U_{avg} = 10 kJ/cm^3 to ≈ 5.2 at 40 kJ/cm^3. This trend is due to the rise of the plasma expansion velocity with increasing energy density. The initial plasma expansion velocity, particle velocity behind the shock front and bubble wall velocity are identical [[40,42]. Previous experiments on ps and ns breakdown showed that the initial bubble wall velocity increases from subsonic values (650 m/s) at $U_{avg} \approx 10$ kJ/cm^3 to supersonic values (2100 m/s) at $U_{avg} \approx 40$ kJ/cm^3 [40]. With supersonic plasma expansion, a shock front is formed almost instantaneously, and E_{SW}/E_B becomes very large.

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